Flow-based Framework Solving Multiple Processes PDEs

Anonymous Author(s)
Affiliation
Address
email

Abstract

This paper proposes a novel method for solving multiple processes PDEs using a few-step flow matching with multiple vector fields, which correspond to multiple processes. Each vector field is modeled to depend not only on the corresponding process but also on the other processes, which reflect the interaction between different processes in a complex physics system.

1 Introduction

[TODO]

2 Related Works

[TODO]

3 Method

3.1 Problem Setting

In this section, we provide a simple example of the coupled processes system as well as the multiple processes system. We begin with the coupled processes system with coupled kernels κ_1 and κ_2, coupled variable $v_1(x,t)$ and $v_2(x,t)$ given initial condition $v_1(x,0)$ and $v_2(x,0)$. Given $\mathcal{U}$ and $\mathcal{A}$ as the initial condition space and the prediction space, let T denote the generic operator such that $T : \mathcal{U} \to \mathcal{A}$. Without the knowledge of how these two variables are coupled, to solve for $v_1(x,\tau)$ and $v_2(x,\tau)$, we need two operator T_1 and T_2 such that

$$v_1(x,\tau) = T_1 v_1(x,0) = \int_D \kappa_1\big(x, y, v_1(x,0), v_2(x,0), v_1(y,0), v_2(y,0), \kappa_2\big) v_1(y,0) dy \quad (1)$$

$$v_2(x,\tau) = T_2 v_2(x,0) = \int_D \kappa_2\big(x, y, v_1(x,0), v_2(x,0), v_1(y,0), v_2(y,0), \kappa_1\big) v_2(y,0) dy \quad (2)$$

where $D \subset \mathbb{R}^d$ is a bounded domain. The interacting kernels cannot be directly solved without considering the interference from the other kernel. We can rewrite as

$$v_1(x,\tau) = T_1[v_1(x,0), v_2(x,0); \kappa_1, \kappa_2] \quad (3)$$

$$v_2(x,\tau) = T_2[v_1(x,0), v_2(x,0); \kappa_1, \kappa_2] \quad (4)$$

Using the above formulations, we can easily extend to the multiple-process system, which includes n different processes, such that, with $i \in [1,n]$, the process $i-$ th can be expressed as:

$$v_i(x,\tau) = T_i[v_1(x,0), ..., v_n(x,0); \kappa_1, ..., \kappa_n] \quad (5)$$

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Our idea is to model each process operator T_i by a flow model Ψ_i with average velocity field from r to t as $u_i^\theta(z_1^t, ..z_n^t; r, t)$, with variable z_i^t as immediate representation of flow i-th at time $t \in [0, 1]$, such that

$$z_i^1(x) = v_i(x, 0) \tag{6}$$

$$z_i^0(x) = v_i(x, \tau) \tag{7}$$

Then, the ground truth (target) average velocity from 0 to 1 with $i-$th process as:

$$u_i^{\text{tar}}(z_1^1(x), ..z_n^1(x); 0, 1) = v_i(x, 0) - v_i(x, \tau) \tag{8}$$

3.2 Average Velocity with Coupled PDEs Processes

In this section, we first derive the ground truth average velocity fields in the setting of two processes $v_1(x, t)$ and $v_2(x, t)$ as [Geng et al., 2025]. Let $u_1^{\text{ins}}(z_1^t(x), z_2^t(x), t)$ and $u_2^{\text{ins}}(z_1^t(x), z_2^t(x), t)$ as instant velocity at time t, we have the average velocity field from r to t as

$$(t - r)u_1(z_1^t(x), z_2^t(x), r, t) = \int_r^t u_1^{\text{ins}}(z_1^\tau(x), z_2^\tau(x), \tau)d\tau \tag{9}$$

Differentiating both sides with respect to t, we have:

$$u_1(z_1^t(x), z_2^t(x), r, t) + (t - r)\frac{d}{dt}u_1(z_1^t(x), z_2^t(x), r, t) = u_1^{\text{ins}}(z_1^t(x), z_2^t(x), t) \tag{10}$$

Note that $\frac{dr}{dt} = 0$ and $u_1^{\text{ins}}(z_1^t(x), z_2^t(x), t) = \frac{d}{dt}(z_1^t(x))$ and $u_2^{\text{ins}}(z_1^t(x), z_2^t(x), t) = \frac{d}{dt}(z_2^t(x))$, we have the time derivative of average velocity $u_1(z_1^t(x), z_2^t(x), r, t)$ as:

$$\begin{aligned}\frac{d}{dt}u_1(z_1^t(x), z_2^t(x), r, t) = \partial_t u_1(z_1^t(x), z_2^t(x), r, t) \\ + u_1^{\text{ins}}(z_1^t(x), z_2^t(x), t)\partial_{z_1^t}u_1(z_1^t(x), z_2^t(x), r, t) \\ + u_2^{\text{ins}}(z_1^t(x), z_2^t(x), t)\partial_{z_2^t}u_1(z_1^t(x), z_2^t(x), r, t)\end{aligned} \tag{11}$$

Substitution from the equation 11 to the equation 10, we then have the form of the ground truth (target) average velocity field as:

$$u_1^{\text{tar}}(z_1^t(x), z_2^t(x), r, t) = u_1^{\text{ins}}(z_1^t(x), z_2^t(x), t) - (t - r)\frac{d}{dt}u_1(z_1^t(x), z_2^t(x), r, t) \tag{12}$$

3.3 Average Velocity with Multiple PDEs Processes

In this section, we provide the analysis of the ground truth average velocity field in the setting of multiple processes $v_i(x, t)$ with $i \in [1, n]$. Derive the same step as Section 3.2, we have the ground truth (target) average velocity field for process $i-$th as:

$$u_i^{\text{tar}}(z_1^t(x), .., z_n^t(x), r, t) = u_i^{\text{ins}}(z_1^t(x), .., z_n^t(x), t) - (t - r)\frac{d}{dt}u_i(z_1^t(x), .., z_2^t(x), r, t) \tag{13}$$

with the time derivative of the average velocity $u_i(z_1^t(x), .., z_n^t(x), r, t)$ as:

$$\begin{aligned}\frac{d}{dt}u_i(z_1^t(x), ..., z_n^t(x), r, t) = \partial_t u_i(z_1^t(x), ..., z_n^t(x), r, t) \\ + \sum_{j=1}^{n} u_j^{\text{ins}}(z_1^t(x), ..., z_n^t(x), t)\partial_{z_j^t}u_i(z_1^t(x), ..., z_n^t(x), r, t)\end{aligned} \tag{14}$$

3.4 Model Instant Velocity Field using Gaussian Process

In Section 3.2 and Section 3.3, we provided the analysis to compute the ground truth average velocity field in both coupled processes and multiple processes settings. However, we still need to estimate the instant velocity field u_i^{ins} to compute the target average velocity effectively. In this section, we introduce the framework to compute this instant velocity based on the Gaussian Process Rasmussen and Williams [2006] in both coupled processes (Section 3.4.1) and multiple processes settings (Section 3.4.2).

3.4.1 Coupled PDE setting

In this section, we will provide the analysis in the coupled PDEs setting, where there are only two processes v_1 and v_2 that interact with each other. Formally, we have the dataset that only includes the initial condition $v_i(x, 0)$ and status $v_i(x, \tau)$ in time τ. We model the dependence between two random variables v_1 and v_2 and time t using Gaussian Process, such that, changing the timescale to flow-based $[0, 1]$, we have the N training pairs $\{(0, z_1^0(x)); z_2^0(x)\}$ and $\{(1, z_1^1(x)); z_2^1(x)\}$. The Gaussian process [Rasmussen and Williams, 2006] models the joint distribution between any finite number of random variables as a joint Gaussian distribution. We denote y as z_2, x as vector of $(t, z_1) \in \mathcal{X} \subset \mathbb{R}^{d+1}$, $\mathrm{y} = f(\mathrm{x}) \in \mathbb{R}^d$, and learn a vector-valued GP [Alvarez et al., 2012], such that:

$$\mathbf{f} \sim \mathcal{GP}(\mathbf{m}, \mathbf{K}) \tag{15}$$

where $\mathbf{m} \in \mathbb{R}^d$ is the mean function and $\mathbf{K}$ is a positive matrix valued function that $(\mathbf{K}(\mathrm{x}, \mathrm{x}'))_{p,q}$ in the matrix $\mathbf{K}(\mathrm{x}, \mathrm{x}')$ correspond to the covariances between the outputs $f_p(\mathrm{x})$ and $f_q(\mathrm{x}')$, which are value of $f(\mathrm{x})$ and $f(\mathrm{x}')$ at dimension p and q. For a set of inputs $\mathbf{X}$, the prior distribution over the vector $\mathbf{f}(X)$ is given by

$$\mathbf{f}(X) \sim \mathcal{N}\big(\mathbf{m}(X), \mathbf{K}(X, X)\big) \tag{16}$$

The predictive distribution for new vector x_* is

$$\mathbf{f}(\mathrm{x}_*) \mid \mathbf{f}, \mathrm{x}_*, X \sim \mathcal{N}\big(\mathbf{f}_*(\mathrm{x}_*), \mathbf{K}_*(\mathrm{x}_*, \mathrm{x}_*)\big) \tag{17}$$

(Noise-free version, we can modify based on the experimental results) with

$$\mathbf{f}_*(\mathrm{x}_*) = \mathbf{K}_{\mathrm{x}_*}^T \big(\mathbf{K}(X, X)\big)^{-1} \mathbf{y} \tag{18}$$

$$\mathbf{K}_*(\mathrm{x}_*, \mathrm{x}_*) = \mathbf{K}(\mathrm{x}_*, \mathrm{x}_*) - \mathbf{K}_{\mathrm{x}_*} \big(\mathbf{K}(X, X)\big)^{-1} \mathbf{K}_{\mathrm{x}_*}^T \tag{19}$$

where $\mathbf{K}_{\mathrm{x}_*} \in \mathbb{R}^{d \times Nd}$ has entries $(\mathbf{K}(\mathrm{x}_*, \mathrm{x}_j))_{p,q}$ for $j \in [1, N]$ and $p, q \in [1, d]$. Thus, given new input $\mathrm{x}_* = (t, z_1)$, we can predict $z_2(\mathrm{x}_*)$ as

$$z_2(\mathrm{x}_*) = \mathbf{f}_*(\mathrm{x}_*) \tag{20}$$

We can also compute the instant velocity of the random variable z_2 at time t as

$$u_2^{\text{ins}}(z_2, \mathrm{x}_*) = \frac{d}{dt}\Big(z_2(\mathrm{x}_*)\Big) = \frac{d}{dt}\Big(\mathbf{f}_*(\mathrm{x}_*)\Big) \tag{21}$$

Note that the derivative of a Gaussian process is another Gaussian process Rasmussen and Williams [2006]. Thus we can use GPs to make predictions about the instant velocity based on derivative information. Given predefined covariance function $k(\cdot, \cdot)$, we have covariance between function values and partial derivatives as

$$\begin{aligned} \operatorname{cov}\Big(\mathbf{f}_*(\mathrm{x_i}), \frac{\partial \mathbf{f}_*(\mathrm{x_j})}{\partial t}\Big) &= \frac{\partial k(\mathrm{x}_i, \mathrm{x}_j)}{\partial t} \\ \operatorname{cov}\Big(\frac{\partial \mathbf{f}_*(\mathrm{x_i})}{\partial t}, \frac{\partial \mathbf{f}_*(\mathrm{x_j})}{\partial t}\Big) &= \frac{\partial^2 k(\mathrm{x}_i, \mathrm{x}_j)}{\partial t} \end{aligned} \tag{22}$$

For practical implementation, we define the input covariance function as the squared-exponential covariance function as

$$k(\mathrm{x}_i, \mathrm{x}_j) = \sigma_f^2 \exp\Big(-\frac{1}{2\ell^2}\|\mathrm{x}_i - \mathrm{x}_j\|_2^2\Big) \tag{23}$$

where ℓ is the length-scale and σ_f^2 is the signal variance. To encode the interaction among the output dimensions, we use separable kernels and sum of separable kernels (SOS kernels) [Alvarez et al., 2012] as the output kernel function, such that

$$\big(\mathbf{K}(\mathrm{x}_i, \mathrm{x}_j)\big)_{p,q} = k(\mathrm{x}_i, \mathrm{x}_j) k_T(p, q) \tag{24}$$

where k, k_T are scalar kernels on $\mathcal{X} \times \mathcal{X}$ and $\{1, ..., n\} \times \{1, ..., n\}$. Thus, we have the matrix expression as

$$\mathbf{K}(\mathrm{x}_i, \mathrm{x}_j) = k(\mathrm{x}_i, \mathrm{x}_j)\mathbf{B} \tag{25}$$

$$\mathbf{K}(X, X) = \mathbf{B} \otimes k(X, X) \tag{26}$$

where $\mathbf{B}$ is a $d \times d$ symmetric and positive semi-definite matrix, which can be learned via maximizing the log-likelihood [Bonilla et al., 2007]. So that, given $\mathrm{x}_* = (t, z_1)$ we can sample the value of $z_2(\mathrm{x}_*)$ and also compute the instant velocity $u_2^{\text{ins}}(z_2, \mathrm{x}_*)$, a lasting bottleneck is how we can sample (t, z_1). Inspired from Dice Strategy [Xiao et al., 2023] to randomly decide the interaction order between each process in a physical system in the current layer. We first sample $t \sim U(0, 1)$ from a uniform distribution, and we then deploy the dice strategy to randomly decide the interaction order between processes v_1 and v_2 with a predefined probability $p \in [0, 1]$. If v_1 interact to v_2, we model the value of z_1^t as the linear interpolant of time with a fixed ending point

$$z_1^t = (1 - t)z_1^0 + tz_1^1 \tag{27}$$

Then, we can plug $\mathrm{x}_* = (t, z_1^t)$ into equation 20 and 21 to compute z_2^t as well as its instant velocity field.

3.4.2 Multiple PDE setting

In this section, we will provide the analysis of multiple PDE settings, where there are n processes $v_1, ..., v_n$ that interact with each other. To extend the coupled PDE setting to multiple PDE setting, the simple idea is to construct multiple GP mappings: $(t, z_i) \to z_j$ with $i, j \in [1, n]$ and $i \neq j$, and using the Dice Strategy to randomly decide the interaction order between processes. The main bottleneck of this approach is the computational cost, as we have to construct $n!$ GP mappings for n processes. To address this problem, we propose a hybrid approach that combines the linear interpolant and GP mappings randomly in each time period t. For instance, we first sample time $t \sim U(0, 1)$, then randomly select $\mathcal{A}(t) = \{i_c, j_c\}_{c=1}^{k(t)}$ interaction pairs set to construct $k(t)$ GP mappings. The leftover processes are inferred by the linear interpolants of time, such that

$$z_i^t = (1 - t)z_i^0 + tz_i^1 \tag{28}$$

3.5 Training Algorithm

In this section, we provide our training algorithm using the flow-based model to solve coupled PDE and multiple PDE with the average velocity derived in Section 3.2 and Section 3.3. We first denote $Z(t) = (z_1^t, z_2^t, ..., z_n^t)$ and introduce the global average velocity loss for each process i as

$$\mathcal{L}_{\text{global}}^i(\theta) = \|u_i^\theta\big(Z(1), 0, 1\big) - (z_i^1 - z_i^0)\|_2^2 \tag{29}$$

We note that the average velocity fields not only need to satisfy the global constraints, but also need to satisfy the local constraints, which are derived from their definitions at each time t. Let $u_i^{\text{tar}}(Z(t), r, t)$ be the target average velocity field from Eq. 12 for coupled PDE setting and Eq. 13 for multiple PDE setting. We have the local velocity loss for each process i as

$$\mathcal{L}_{\text{local}}^i(\theta, r, t) = \|u_i^\theta\big(Z(t), r, t\big) - \text{sg}\big(u_i^{\text{tar}}(Z(t), r, t)\big)\|_2^2 \tag{30}$$

In the local loss function, a stop-gradient (sg) operation is applied on the target average velocity fields $u_i^{\text{tar}}(Z(t), r, t)$ following the common practice [Geng et al., 2025] to avoid double backpropagation. Given the dataset as $\mathcal{D}$, we have the combined loss function as

$$\mathcal{L}(\theta, r, t) = \mathbb{E}_{\mathcal{D}}\Big[\sum_{i=1}^{n} \mathcal{L}_{\text{global}}^i(\theta)\Big] + \lambda \mathbb{E}_{\mathcal{D}}\Big[\sum_{i=1}^{n} \mathcal{L}_{\text{local}}^i(\theta, r, t)\Big] \tag{31}$$

The detailed training and sampling algorithm are presented in Alg. 1 and Alg. 2.

4 Experiment Results

[TODO]

5 Discussion

[TODO]

6 Conclusion

[TODO]

Algorithm 1 Training Algorithm

Input: Dataset $\mathcal{D}$, number of epochs n_0.
Output: The flow model weight θ.
for $i = 1$ to n_0 **do**
    `t,r = sample_t_r()`
    Sample $Z(t) = (z_1^t, ..., z_n^t)$ using $\mathcal{GP}$ (Section 3.3)
    Compute $\mathcal{L}(\theta, r, t)$ as Eq. 31
    Update θ via minimizing $\mathcal{L}(\theta, r, t)$
end for
Return the weight θ

Algorithm 2 One-Step Sampling Algorithm

Input: Test point $Z(1) = (z_1^1, ..., z_n^1)$, pretrained weight θ
for $i = 1$ to n **do**
    $z_i^0 = z_i^1 - u_i^\theta(Z(1), 0, 1)$
end for
$Z(0) = (z_1^0, ..., z_n^0)$
Return $Z(0)$

A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

Note: Think of the appendix as "optional reading" for reviewers. The paper must be able to stand alone without the appendix; for example, adding critical experiments that support the main claims to an appendix is inappropriate.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

References

Mauricio A. Alvarez, Lorenzo Rosasco, and Neil D. Lawrence. Kernels for vector-valued functions: a review, 2012. URL `https://arxiv.org/abs/1106.6251`.

Edwin V Bonilla, Kian Chai, and Christopher Williams. Multi-task gaussian process prediction. In J. Platt, D. Koller, Y. Singer, and S. Roweis, editors, *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc., 2007. URL `https://proceedings.neurips.cc/paper_files/paper/2007/file/66368270ffd51418ec58bd793f2d9b1b-Paper.pdf`.

Zhengyang Geng, Mingyang Deng, Xingjian Bai, J Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL `https://openreview.net/forum?id=uWj4s7rMnR`.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. The MIT Press, 2006.

Xiongye Xiao, Defu Cao, Ruochen Yang, Gaurav Gupta, Gengshuo Liu, Chenzhong Yin, Radu Balan, and Paul Bogdan. Coupled multiwavelet operator learning for coupled differential equations. In *The Eleventh International Conference on Learning Representations*, 2023. URL `https://openreview.net/forum?id=kIo_C6QmMOM`.